VAIBHAV KULSHRESTHA

DATA SCIENTIST | MACHINE LEARNING | GENERATIVE AI | COMPUTER VISION

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with 5+ years of experience spanning production machine learning, computer vision, deep learning, generative AI, agentic systems, and cloud engineering. Hands-on record of delivering ML systems for railway safety, video analytics, biomedical research, NLP, and enterprise automation, with strengths across Python, SQL, Azure, model pipelines, RAG, and productionization. Experienced in translating complex technical problems into scalable, reliable AI solutions. Canadian work history includes graduate research at Princess Margaret Cancer Centre.

TECHNICAL COMPETENCIES

Programming: Python, SQL, C++
ML / AI: Deep Learning, Computer Vision, Transformers, LLMs, RAG, Agentic AI, Time Series, Transfer Learning, Clustering
Data Engineering: ML Pipelines, Data Processing, Model Evaluation, MLOps, Kubeflow, Docker, CI/CD, MySQL
Cloud / Platforms: Microsoft Azure, Azure DevOps, Raspberry Pi / Edge AI
Visualization / Analytics: Video Analytics, Operational Analytics, Data-driven Monitoring

PROFESSIONAL EXPERIENCE

Optimus Information Inc. — Software Engineer | Noida, India | Jun 2026 – Present

- Built an AI software architect that converts complex requirements into structured requirements, solution architectures, Azure recommendations, delivery plans, and technical designs, including architecture assessment against organizational standards.
- Automated the end-to-end clinical workflow from CMS ingestion to final patient delivery, eliminating manual data entry and accelerating report turnaround times.
- Designed and built an agentic ITSM platform on Azure DevOps using the Microsoft Agent Framework, RAG, enterprise knowledge, and workflow automation to accelerate service delivery.

Wi-Tronix — Data Scientist | Bolingbrook, IL | Feb 2024 – Jun 2026

- Led production computer vision systems for railway safety, including object detection for trespasser detection and track segmentation from moving locomotives.
- Designed computer vision solutions for infrastructure inspection, crew-behaviour monitoring, and passenger-occupancy estimation under motion blur, changing viewpoints, and variable illumination.
- Built scalable ML data pipelines and model workflows using Python and SQL, supporting repeatable training, evaluation, and deployment across multiple products.
- Created reusable, versioned Python packages with testing and documentation, establishing engineering standards and improving maintainability and cross-team adoption.

Slytek, Inc. — Lead AI Engineer | Denver, CO | Jun 2023 – Feb 2024

- Integrated Raspberry Pi 4B and Camera Module 3 for real-time video analysis to monitor operator compliance with standard operating procedures.
- Developed CI/CD for a C++ application using unit testing, Azure DevOps, and Docker to support repeatable integration and deployment.
- Integrated Raspberry Pi 4 with ADE9000 to capture and analyze voltage, current, and power measurements for data monitoring.

Arizona State University — Graduate Services Assistant | Tempe, AZ | Mar 2022 – Dec 2022

- Supported teaching and grading for graduate-level deep learning courses, including CSE 598: Introduction to Deep Learning and CSE 598: Intro to Deep Learning in Visual Computing.

SemiCab — Machine Learning Intern | Atlanta, GA | May 2022 – Aug 2022

- Built an NLP workflow to parse and classify operations emails by intent and extract relevant information; applied ULMFiT transfer learning, transformer-based zero-/few-shot learning, and DBSCAN clustering.

Drishti — Machine Learning Engineer / Software Engineer Intern | Bengaluru, India | Jan 2021 – Aug 2021

- Developed a Kubeflow pipeline using a Boundary Matching Network, increasing temporal proposal accuracy by 5%.
- Optimized a streaming pipeline, cutting end-to-end per-frame latency by 50%, and migrated deep-learning model operations to a more efficient API.

SELECTED RESEARCH & EARLIER EXPERIENCE

Princess Margaret Cancer Centre — Master Thesis Project | Toronto, ON | Jun 2020 – Dec 2020

- Identified isoforms and isoform switches as candidate biomarkers for Acute Myeloid Leukemia (AML), supporting research into early diagnosis and treatment strategies.
- Applied parallel processing to optimize computational scripts, reducing execution time and improving computational efficiency.

Birla Institute of Technology and Science, Pilani — Student Research Assistant | Goa, India | Oct 2019 – Mar 2020

- Applied AI to dengue virus genomic data to predict infection severity; analyzed RNA secondary-structure prediction tools and developed a tool addressing identified shortcomings.

- Published research on dengue outbreak/severity prediction and RNA secondary-structure prediction tools.

Optimus Information Inc. — Software Engineer Intern | Noida, India | Aug 2019 – Oct 2019
- Applied classification and regression algorithms in Golang and developed two APIs for processing data stored in MySQL.

University of Tokyo, Lab of Functional Analysis in silico — Summer Research Intern | Tokyo, Japan | May 2018 – Jul 2018
- Applied non-negative matrix factorization to molecular and genomic datasets to identify anticancer drug relationships and tumour-suppressor genes.
- Developed and published the BigNMF Python package on PyPI, reaching 46,000+ downloads.

Prokure — Software Engineer Intern | New Delhi, India | May 2017 – Jul 2017
- Automated third-party order delivery, built a web-based administration panel, and web-scraped e-commerce inventory into a MySQL database.

EDUCATION

Arizona State University — M.S., Robotics and Autonomous Systems (AI) | Aug 2021 – May 2023

Birla Institute of Technology and Science, Pilani – Goa Campus — B.E., Electrical and Electronics Engineering | Aug 2015 – Aug 2021

Birla Institute of Technology and Science, Pilani – Goa Campus — M.S., Biological Sciences | Aug 2015 – Aug 2021

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AZ-900
- Microsoft Copilot: The Art of Prompt Writing
- SAFe 6 course completion
- Wi-Tronix and Rail Energy Management
- Python (Basic) Certificate
- CodeChef Foundation Level DSA Exam — Problem Solving (Basic)

LANGUAGES

English (Full Professional) | Hindi (Native/Bilingual) | French (Limited Working) | Japanese (Elementary)

PUBLICATIONS

- Dengue outbreak and severity prediction: current methods and the future scope [link]
- A Brief Review and Comparative Analysis of RNA Secondary Structure Prediction Tools [link]